Shrivatsa Deshmukh

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Summary: *Computational neuroscience and NeuroAI researcher with a background in physics. Builds and tests models of neural data from single neurons to whole-brain EEG, with a focus on interpretability and reliable inference: what models learn, and whether their explanations can be trusted.*

SKILLS

Neuro & Computational: Neural Mass Models, Simulation-Based Inference (Normalizing Flows), Bayesian & Hierarchical Models, Model Interpretability, EEG, fMRI, Calcium Imaging, Electrophysiology (spike data)

Programming & ML: Python, NumPy, SciPy, Pandas, PyTorch, Scikit-learn, MATLAB, Git, Linux/Bash, MNE-Python

PUBLICATIONS

- **Shrivatsa Deshmukh**, Sasha Brenner, Nico Scherf, and Thomas R. Knösche. "Gate-Based Feature Importance in Simulation-Based Inference for EEG: A Cautionary Tale." *NeurIPS Workshops 2026 - Interp4Discovery and Sim2Science.* doi:10.48550/arXiv.2610.30364
- **Shrivatsa Deshmukh**, Vitthal Khatik, and Anupam Saxena. "Robust Fusion Model for Handling EMG and Computer Vision Data in Prosthetic Hand Control." *IEEE Sensors Letters 2023*. doi:10.1109/LSENS.2023.3301837

EXPERIENCE

Noel Lab, University of Minnesota | *Research Assistant* *May 2026 – December 2026*

- Built a trial-level psychometric model linking pre-stimulus population activity to prior use, sensitivity and lapses, and applied it to Neuropixels recordings from wild-type and three autism-model mouse lines (*Cntnap2*, *Fmr1*, *Shank3B*; Noel et al., 2025) and 53 independent WT mice from the IBL Brain-Wide Map.
- Found that in autism-model mice, lapses became less likely as pre-stimulus activity in the cerebral nuclei rose, unlike in WT (corrected $p = 0.039$), although lapse rates did not differ; the effect vanished under time-shifted controls and was not explained by task engagement.

Alfonso Araque Lab, University of Minnesota | *Research Assistant* *January 2026 – May 2026*

- Analyzed AQuA2-segmented astrocyte calcium imaging (3,008 events, 48 recordings) to test how psilocybin alters astrocyte signaling in wild-type (WT), 5-HT_{2A}-blocked, and IP_3R2 conditional-knockout slices.
- Found that psilocybin increased astrocyte integrated calcium signal by 25% and event duration by 20% (both $p < 0.001$); both effects were abolished in IP_3R2 knockouts ($p < 0.01$ vs. WT), with a non-significant trend toward attenuation under 5-HT_{2A} blockade.

Brain Networks Group, Max Planck Institute CBS | *Guest Researcher* *May 2025 – September 2025*

- Built a simulation-based inference pipeline (neural posterior estimation with a neural spline flow) to infer physiological parameters of a Jansen-Rit neural mass model from EEG simulated through a realistic head-model forward projection; R^2 = 0.79 with well-calibrated posteriors.
- Showed that learned feature gates are not a reliable readout of feature importance: though stable across seeds and accuracy-neutral, they anti-correlated with retrain-based ablation importance (Spearman $\rho \approx -0.6$) and down-weighted the most informative EEG feature.

Indian Institute of Technology Kanpur | *Machine Learning Researcher* *January 2023 – August 2023*

- Designed an EMG–vision fusion model for prosthetic-hand grasp classification that uses object-detection output (YOLOv5) as a classifier input, making it robust to detection errors.
- Improved accuracy by up to 11.57% over prior fusion methods and, unlike them, stayed above EMG-only accuracy under 25% object-label corruption.

PROJECTS

What Does POYO Learn About Neurons? | GitHub

- Probed POYO-MP, a pretrained transformer foundation model for neural decoding, without retraining; showed its neuron embeddings encode each neuron's tuning and can be reused on other recording days, recovering 91% of accuracy without labeled data.
- Tested whether neurons share a "vocabulary" like language-model tokens; found they did not reduce to a few types, but a shared 16-dimensional space kept 96% of accuracy and transferred across monkeys.

Neural Phoneme Decoder for Speech BCI | GitHub

- Reproduced the gated recurrent unit (GRU) phoneme decoder from Willett et al. (2023, Nature), isolating the decoder stage from the full Kaldi pipeline.
- Reduced compute requirements via hyperparameter optimization to enable single-GPU training, achieving 78.77% phoneme accuracy within 1.5% of the published benchmark.

EDUCATION

University of Minnesota – Twin Cities *August 2024 – December 2026*
Master of Science in Medical Robotics **GPA: 3.95 / 4.00**

Birla Institute of Technology and Science (BITS), Pilani *August 2018 – May 2023*
Master of Science in Physics &
Bachelor of Engineering in Mechanical Engineering **GPA: 8.37 / 10.00**